Sahaj Bhadja

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Summary

Software engineer with 2+ years building **Python/C++-adjacent systems**, **test-oriented APIs**, and **cloud workflows**. Strong in debugging, multithreading, CI/CD, and object-oriented software across production and research environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Architected **Azure SQL and Blob services** for 500+ users, tuning indexed queries to cut dashboard latency from **1.2 seconds to 350 milliseconds** across production workflows.
- Built **fault-tolerant job execution** for 1,000+ monthly workflows with retries and isolation, improving throughput by **3x** while strengthening debugging visibility under concurrent load.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Designed **concurrent batch-processing pipelines** for 1,000+ uploaded files using multithreading, REST APIs, and AWS RDS-backed services, reducing turnaround time by **60%** during peak execution.
- Engineered **Flask services** and React workflows handling **10,000+ daily requests**, integrating secure ingestion and CI/CD-backed releases for production instructor-facing operations.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **transformer evaluation pipelines** across **100K+ trajectory logs**, enabling reproducible benchmarking, experiment tracking, and failure analysis for long-horizon planning models in simulation-heavy environments.
- Architected **map-aware encoders** with spatial-temporal attention, improving closed-loop planning stability by **15%** while strengthening systems-level debugging and model validation rigor.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **cross-platform API and ETL microservices** with SQL, AWS Glue, and S3, improving response times by **30%** while supporting validation and operational reliability across shared environments.
- Maintained **Jenkins CI pipelines** and 20+ automated suites with PyTest and Mockito, increasing release reliability by **68%** while improving debugging speed and regression control.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **MERN APIs and WebSocket modules** supporting **2M+ synchronized records**, improving consistency across real-time modular workflows and cross-service coordination.
- Automated **Docker-based build and test workflows** with Jenkins CI/CD, cutting release time by **50%** while improving deployment consistency and staged validation across environments.

Skills

Languages: **C++**, **Python**, Java, TypeScript, JavaScript, SQL, Bash; **Software & Test**: Flask, FastAPI, React, REST APIs, WebSockets, microservices, multithreading, object-oriented design, debugging, performance optimization, automated testing
Platforms & Tools: Windows, Linux, macOS, Git, Docker, Jenkins, CI/CD, PyTest, Mockito, AWS, Azure, Azure SQL, AWS RDS, S3, AWS Glue, SQL Server, PostgreSQL, MySQL, MongoDB, logging, monitoring, Agile/Scrum

Projects

CPU-scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2*
- Implemented **FCFS, SJF, SRT, RR, priority scheduling, and MLQ** in C++ with interactive controls, EJ-2 Gantt visualization, and dynamic waiting-time calculations across 1K+ simulated process combinations.

Linguistix | *Python, AWS, Docker, CI/CD*
- Developed a containerized **Python REST API** on AWS with automated CI/CD releases and validation checks, achieving **87% accuracy** while supporting reliable production-style data-processing workflows.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Worked across **Java and TypeScript codebases** with Docker, Maven, Node.js, and Yarn to support cross-platform database workflows on Windows, macOS, and Linux with AI-ready local deployment.